



Fully funded PhD and Postdoc positions at Nanyang Technological University, Singapore

I will be joining Nanyang Technological University (NTU), Singapore, as a tenure-track Assistant Professor in the Department of Civil and Environmental Engineering in Fall 2026. I am honored to be appointed as a Nanyang Assistant Professor, one of NTU's prestigious faculty appointments. I am recruiting multiple fully funded PhD and postdoctoral positions to advance research in civil infrastructure monitoring, distributed acoustic sensing, urban geophysics, and AI/ML for urban resilience. Our goal is to develop scalable and cost-effective approaches for monitoring the built and natural environment to enhance safety, sustainability, and resilience.

What we offer

- *PhD students*: Funding covers tuition, stipend, and benefits, with additional support for visiting scholar opportunities at universities worldwide.
- *Postdoctoral researchers*: Initial one-year appointment, extendable up to three years based on performance.
- *Research assistants and visiting scholars* are also welcome.
- Funding available for travel, training, computing resources, and publications.
- Competitive packages, access to state-of-the-art equipment and facilities, and collaborations with world-class researchers.

Research Topics

1. Civil Infrastructure Monitoring Using Distributed Sensing (PhD and Postdoc)

The successful candidate will conduct research on the scalable monitoring of civil infrastructure systems, such as bridges, railroads, and roadway structures, using distributed sensing technologies, with a particular emphasis on distributed acoustic sensing (DAS) on pre-existing telecom fiber networks and mobile sensing using vehicle fleets.

This effort will involve developing algorithms to analyze guided waves, dynamic and static responses, sensor and data fusion, and system identification, as well as planning and executing both lab-scale and field experiments to investigate structural responses captured by distributed and multi-modal sensors.

References [1–4] provide related publications for this research topic.

2. Urban Geophysics and Seismic Noise Monitoring (only PhD)

The successful candidates will conduct research in urban and environmental geophysics using DAS on pre-existing telecom fiber networks. The goal is to harness ambient and anthropogenic seismic noise for urban subsurface characterization and environmental monitoring, including geohazard detection such as sinkhole and subsidence. Core tasks will include developing algorithms for passive seismology, geophysical imaging, and seismic source characterization, as well as designing and carrying out field experiments in dense urban environments.

References [5–8] provide related publications for this research topic.

3. AI/ML for Urban Resilience and Smart Infrastructure (only PhD)

The successful candidate will apply artificial intelligence and machine learning to infrastructure assessment, disaster resilience, and urban system monitoring. Potential directions include 1) building foundation models for sensor data embedding and downstream tasks, 2) developing generative and agent-based models for pre-disaster simulation and post-disaster decision support, 3) advancing domain adaptation and transfer learning methods for infrastructure assessment and diagnostics, and 4) creating multimodal data fusion frameworks that integrate distributed fiber-optic sensing, remote sensing, and mobility data.

References [3,9] provide related publications for this research topic.

Qualifications

- *PhD students*: Bachelor's or Master's in Civil/Structural Engineering, Engineering Mechanics, Seismology, Geophysics, Computer/Data Science, Electrical/Computer Engineering, Applied Math/Physics, or a related field. Strong motivation, effective communication, and a willingness to engage in fieldwork are essential. For NTU's PhD program details and requirements, please refer [here](#).
- *Postdoctoral researchers*: PhD in a related field with a strong research record and expertise in at least one of the following: distributed fiber-optic sensing, structural health monitoring, structural dynamics, seismology, geophysics, or AI/ML for science and engineering.

Preferred Skills

- Experience in LLMs, LVMs, foundation models, or world models is a plus
- Hands-on experience in structural testing, such as modal testing, acoustic emission, or ultrasonic testing
- Experience in distributed fiber-optic sensing (DAS, DSS, DTS, OFDR, multiplex FBG), including analysis and deployment
- Strong numerical modeling and simulation skills (e.g., COMSOL Multiphysics)
- Proficiency in programming (e.g., Python, C++, Julia) and familiarity with AI/ML frameworks (e.g., PyTorch, TensorFlow, scikit-learn)
- Experience with reproducible codebases (e.g., GitHub) and collaborative workflows
- Background in distributed/parallel computing for large-scale seismic or DAS data

How to Apply

Send your CV, selected publications, transcript, and research statement (optional) to jingxiao.liu.jxl@gmail.com with the subject line: "PhD/Postdoc/RA Application."

About the PI (<https://www.jingxiao-liu.com/>)

Jingxiao Liu received his B.S. in Civil Engineering from Central South University, China, his M.S. in Civil Engineering from Carnegie Mellon University, and his Ph.D. in Structural Engineering with a minor in Electrical Engineering from Stanford University. He is currently a Postdoctoral Fellow at the Massachusetts Institute of Technology. His research has been recognized with multiple best paper and presentation awards from ASCE, ASME, SEM, and ACM.

My *advising philosophy* is centered on helping PhD students and postdoctoral researchers accomplish impactful research and grow into independent scholars. For PhD students, my goal is to guide you in producing research that makes a rigorous and impactful contribution to the field. I do not believe that success should be measured solely by the number of papers or journal impact factors. Instead, I emphasize originality, rigor, and long-term contributions. In your early years, I will be hands-on in guiding you through the academic world: developing research ethics, building expertise, identifying the right scientific problems and communities, and strengthening your skills in scientific writing, publishing, and presenting. In your later years, my focus will shift to supporting you in pursuing independent research: shaping your dissertation, preparing for your next career steps, and building confidence in your research vision.

For postdoctoral researchers, I see your role as both collaborators and mentors to students within the group. You will work closely with me, the students, and external collaborators to advance our research agenda while also developing your own independent research vision. I will encourage you to take leadership in projects, proposals, and collaborations, and to use this stage to build the skills and networks that will prepare you for your next faculty or research position.

References

- [1] Liu, J., Aggarwal, J., Hwang, D., Lee, J., Wu, Y., Yin, F., Li, H., Santi, P., Sohn, H., Biondi, B. and Ratti, C., 2025. Bridge Monitoring Using Existing Telecom Fiber-Optic Networks. <https://doi.org/10.21203/rs.3.rs-7603654/v1>
- [2] Liu, J., Yuan, S., Luo, B., Biondi, B. and Noh, H.Y., 2023. Turning telecommunication fiber-optic cables into distributed acoustic sensors for vibration-based bridge health monitoring. *Structural Control and Health Monitoring*, 2023(1), p.3902306.
- [3] Liu, J., Xu, S., Bergés, M. and Noh, H.Y., 2023. HierMUD: Hierarchical multi-task unsupervised domain adaptation between bridges for drive-by damage diagnosis. *Structural Health Monitoring*, 22(3), pp.1941-1968.
- [4] Liu, J., Yuan, S., Dong, Y., Biondi, B. and Noh, H.Y., 2023. TelecomTM: A fine-grained and ubiquitous traffic monitoring system using pre-existing telecommunication fiber-optic cables as sensors. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(2), pp.1-24.
- [5] Liu, J., Li, H., Noh, H.Y., Santi, P., Biondi, B. and Ratti, C., 2025. Urban sensing using existing fiber-optic networks. *Nature Communications*, 16(1), p.3091.
- [6] Liu, J., Li, H., Yuan, S., Noh, H.Y. and Biondi, B., 2025. Characterizing Vehicle-Induced Distributed Acoustic Sensing Signals for Accurate Urban Near-Surface Imaging. *Seismological Research Letters*.

- [7] Li, H., Liu, J. and Biondi, B.L., 2025. Near-surface imaging and monitoring enabled by urban distributed acoustic sensing seismic arrays. *The Leading Edge*, 44(8), pp.588-597.
- [8] Yuan, S., Liu, J., Noh, H.Y., Clapp, R. and Biondi, B., 2024. Using vehicle-induced DAS signals for near-surface characterization with high spatiotemporal resolution. *Journal of Geophysical Research: Solid Earth*, 129(4), p.e2023JB028033.
- [9] Li, L., Li, D., Ou, Z., Xu, X., Liu, J., Ma, Z., Yu, R. and Deng, M., 2025. LLMs as World Models: Data-Driven and Human-Centered Pre-Event Simulation for Disaster Impact Assessment. (EMNLP 2025) *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.